

Rate-Quantified BNT Cross-Overlap Decay on the **SectorBNT** Sector Surface

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1 Notation

Fix a sector decomposition P with basis blocks $\{P.\mathbf{basis}\ j\}_{j=0}^{g-1}$ and a spectral level $\lambda_j \in \mathbb{C} \setminus \{0\}$ with $\|\lambda_0\| > \|\lambda_1\| > \dots > \|\lambda_{g-1}\|$ and $\|\lambda_0\| = 1$. The bilinear cross-overlap between basis blocks j and k at length N is

$$O_{j,k}(N) := \text{mpvOverlap}(P.\mathbf{basis}\ j, P.\mathbf{basis}\ k)(N) = \sum_{\sigma} \text{mpv}(P.\mathbf{basis}\ j)(\sigma) \overline{\text{mpv}(P.\mathbf{basis}\ k)(\sigma)}, \quad (1)$$

written `mpvOverlap` in the formalization.¹

For every ordered pair $j \neq k$ we are interested in a geometric bound

$$\|O_{j,k}(N)\| \leq C_{j,k} \eta_{j,k}^N \quad (\forall N \in \mathbb{N}), \quad (2)$$

with $C_{j,k} \geq 0$ and a rate $\eta_{j,k} \in [0, 1)$.

2 Cited assertion

The Step 1 argument in the proof of Theorem `thm1` [CPGSV16, line 1182] relies on $O_{j,k_0}(N) \rightarrow 0$ for $j \neq k_0$ to discard cross-overlap terms when projecting an assembled BNT identity onto a distinguished basis block $V^{(N)}(P.\mathbf{basis}\ k_0)$. The qualitative limit is implicit in the BNT structure: distinct normal blocks satisfy

$$O_{j,k}(N) \xrightarrow[N \rightarrow \infty]{} 0 \quad (j \neq k). \quad (3)$$

The argument as written does not quantify the rate of decay in (3).

Definition 4.2 of [CPGSV21] packages (3) indirectly under *eventual linear independence* of the basis MPVs and does not state a rate. The formalization records this eventual linear-independence input in the shared sector-decomposition infrastructure,² while (3) is provided pointwise by `MPSTensor.mpvOverlap_tendsto_zero_of_not_gaugePhaseEquiv_cast_left_of_irreducible_TP`.³

¹Declaration `MPSTensor.mpvOverlap` in `TNLean/MPS/Overlap/Basic`.

²`TNLean/MPS/SharedInfra/SectorDecomposition`.

³`TNLean/MPS/Overlap/CastDecay`.

Neither source equips $\eta_{j,k}$ with a quantitative relation to the spectral weight ratios.

3 The retired rate hypothesis

An earlier formal route introduced an explicit rate hypothesis on a sector decomposition P and a chosen spectral level $\lambda : \text{Fin } P.\text{basisCount} \rightarrow \mathbb{C}$. That route has since been retired together with the per-block-projection layer. The mathematical hypothesis it was trying to isolate remains the following: there should exist non-negative families C, η such that for every $j \neq k$,

$$0 \leq C_{j,k}, \quad 0 \leq \eta_{j,k} < 1, \quad \eta_{j,k} \cdot \|\lambda_j / \lambda_k\| < 1, \quad (4)$$

together with the geometric bound (2) for all N .

The surviving `SectorBNT` surface does not package such constants. It keeps the raw BNT weights in `SectorDecomposition` and records qualitative consequences in `TNLean/MPS/FundamentalTheorem/SectorBNT/Api`: the coefficient bound `SectorDecomposition.norm_coeff_le_copies_of_norm_weight_le_one`, within-family cross-decay `MPSTensor.IsBNTCanonicalForm.cross_overlap_basis_tendsto_zero`, the combined-family linear-independence criterion `MPSTensor.IsBNTCanonicalForm.combined_family_eventually_l` and the non-vanishing coefficient statement `MPSTensor.IsBNTCanonicalForm.coeff_not_tendsto_zero_at_block`. None of these declarations supplies the constants in (4).

4 Where such a hypothesis would be needed

The retired per-block-projection route carried an abstract hypothesis excluding asymptotic cancellation of the assembled overlap against a chosen projection block k_0 . The obstruction analysis in `audits/2026-05-13_cpstv16_ft_path_beta_pr2_5_design.md` shows that for a non-dominant k_0 the discharge requires, after isolating $V^{(N)}(P.\text{basis } k_0)$ on the right and renormalising by $\lambda_{k_0}^N$, that every $j \neq k_0$ contribute a term whose modulus is bounded by

$$\|O_{j,k_0}(N)\| \cdot \|\lambda_j / \lambda_{k_0}\|^N \leq C_{j,k_0} (\eta_{j,k_0} \cdot \|\lambda_j / \lambda_{k_0}\|)^N. \quad (5)$$

The right-hand side of (5) converges to 0 exactly when (4) holds. The qualitative limit (3) alone gives $\eta_{j,k_0} < 1$ but not the comparison $\eta_{j,k_0} \cdot \|\lambda_j / \lambda_{k_0}\| < 1$, which is the content of the new hypothesis.

5 Classification

Open gap. The rate condition (4) is a structural input beyond what [CPGSV16, §II] and [CPGSV21, Definition 4.2] make explicit. Its rate is in principle derivable from the spectral gap of the mixed transfer operator that powers `mpvOverlap_tendsto_zero_of_not_gaugePhaseEquiv_cast_left_of_irreducible_TP`, but the compatibility inequality $\eta_{j,k} \cdot \|\lambda_j / \lambda_k\| < 1$ is not implied by the BNT structure and must be either imposed or established analytically.

The analytic derivation from the transfer matrix spectral gap is the subject of *Option 2* in the design memo `audits/2026-05-13_cpsv16_ft_path_beta_pr2_5_design.md` and is not addressed in this note. The current `SectorBNT` route instead keeps the coefficient comparison and the combined-family linear independence in `TNLean/MPS/FundamentalTheorem/SectorBNT`; any future rate-quantified argument should be stated against that surviving surface, not against the retired per-block-projection layer.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.